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4. A scientist was investigating changes in lactic acid concentration in the blood of two students, Cerys and Blodwen. They both ran on a treadmill at the same speed for the same time.

(a) **Table 4.1** shows the results for Cerys.

Table 4.1

Time (minutes)	Concentration of lactic acid (mg per dm^3)
0	60
10	120
20	180
30	200
40	150
50	110
60	70
70	60

(i) Complete **Graph 4.2** of these results by:

- Plotting the concentration of lactic acid for Cerys [2]
- Joining your plots with a ruler. [1]

The first three plots for Cerys and **all** the plots for Blodwen have been done for you.

(ii) From **Graph 4.2**, describe **one** difference between the results for Cerys and Blodwen. [1]

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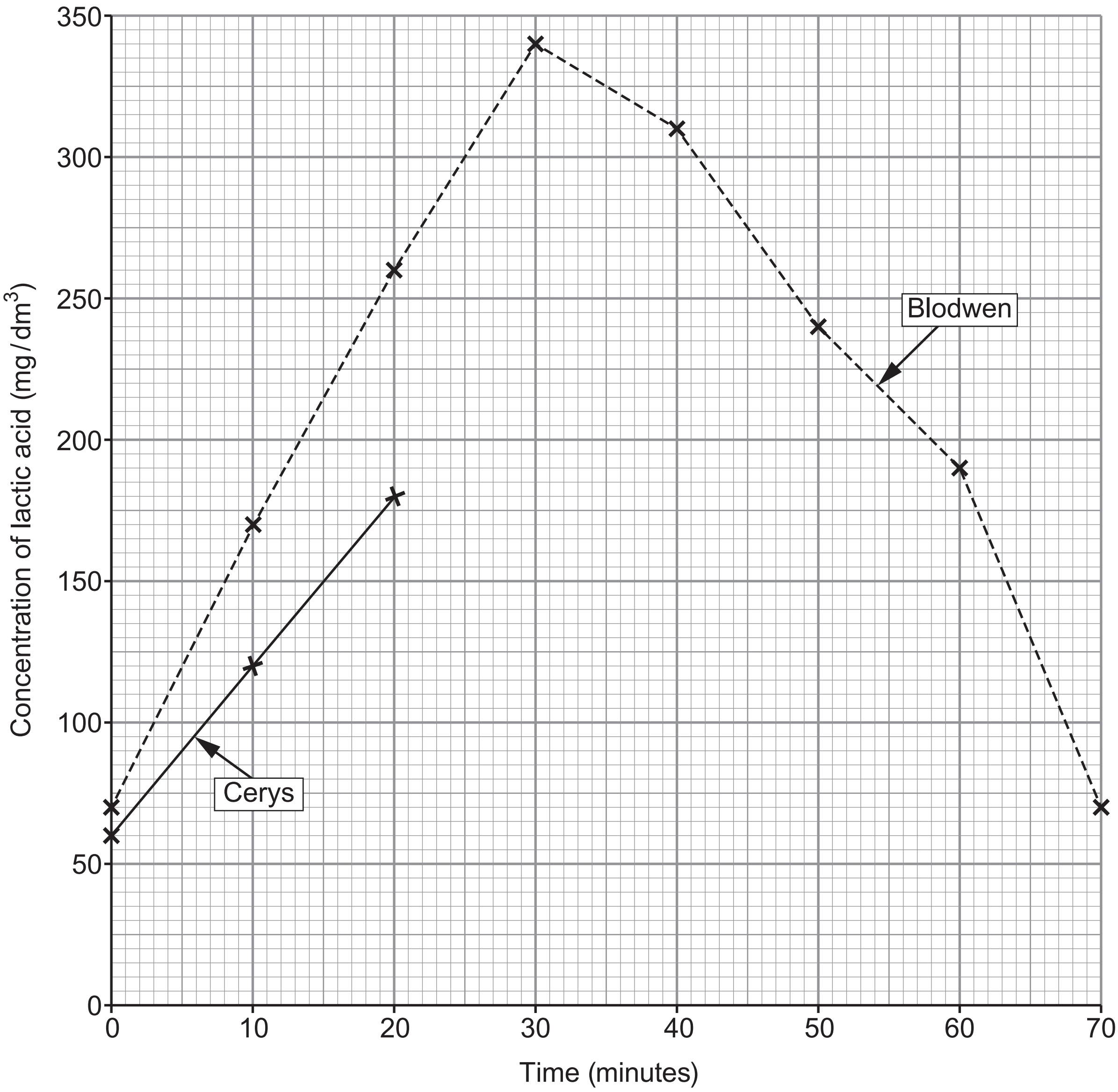
Graph 4.2



(b) Use **Graph 4.2** to:

(i) calculate the maximum increase in lactic acid in the blood for Blodwen; [1]

Increase = ... mg per dm^3

(ii) suggest what time Blodwen stopped running. [1]

Time = ...minutes

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(c) The students are using anaerobic respiration for part of the time they are running. Complete the word equation for anaerobic respiration. [2]

.... ⟶ .... (+ energy)

(d) The students continued to breathe deeply and rapidly after they had finished running. Explain the reason for this. [2]

....

....

....

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5. The amount of pollution in water can be determined by measuring oxygen levels, pH levels or by using indicator species. Polluted water will have low oxygen levels, as bacteria use up oxygen. The water will become more acidic.

Air pollution can be assessed using indicator species, such as lichens. Lichens are organisms that consist of fungi and algae living together. Lichens can grow on tree bark and rocks. There are many species of lichen which can be grouped into three main types: bushy, leafy and crusty. The number of types of lichens present can indicate the level of air pollution.

- Bushy lichens can only survive in unpolluted air.
- Leafy lichens can survive a small amount of air pollution.
- Crusty lichens can survive in more polluted air.
- The more types of lichen growing in an area, the lower the level of air pollution.

(a) Use the information above to **complete Table 5.1** by writing true or false next to each statement. [4]

Table 5.1

Statement	True/False
Bacteria use oxygen for respiration	
Polluted water will be pH 7 (neutral)	
Lichens are plants	
There will be a higher number of different types of lichen in polluted areas	
Bushy lichens can survive in more polluted areas	

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(b) (i) In 2019, a study rated the air quality in the centre of Cardiff as being worse than the centre of London.

Image 5.2 shows the three main types of lichen.

In **Image 5.2 circle the letter (A**, **B**, **C)** of the type of lichen you may expect to find growing in Cardiff city centre if there was a lot of air pollution present. [1]

Image 5.2

A **B** **C**

Bushy Leafy Crusty

A scientist counted the number of species of lichens growing at different distances from the centre of Cardiff. Her results are shown in **Table 5.3**.

Table 5.3

Distance from Cardiff centre (km)	Number of species of lichen
0	2
1	5
2	8
3	10
4	15
5	....
6	26
7	30
8	48
9	56
10	60

(ii) **Complete Table 5.3** by estimating the number of species of lichen likely to be found 5 km from Cardiff centre. [1]

(iii) State the independent and dependent variables in this investigation. [2]

Independent variable ....

Dependent variable ....

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(iv) I. State the relationship between the number of lichen species and the distance from the centre of Cardiff. [1]

....

....

II. Use the information given to explain the relationship between the number of species of lichen and the distance from the centre of Cardiff. [2]

....

....

....

....

(v) State how the scientist could improve her confidence in these results. [1]

....

....

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